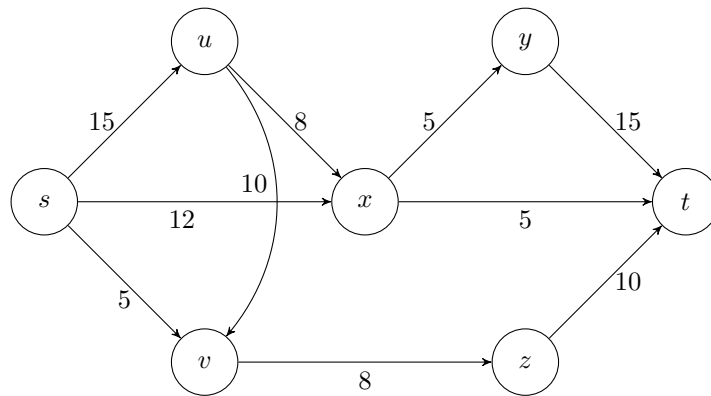


Weekly Assignment 6: Flow algorithms

October 2025

- (15 points) Run the Ford-Fulkerson algorithm on the following flow network. For each iteration, pick the shortest augmenting path from s to t and give the residual network. Finally, give the value of the maximum flow and a minimum cut (S, T) .



- (25 points) Consider the following network (the numbers are edge capacities).

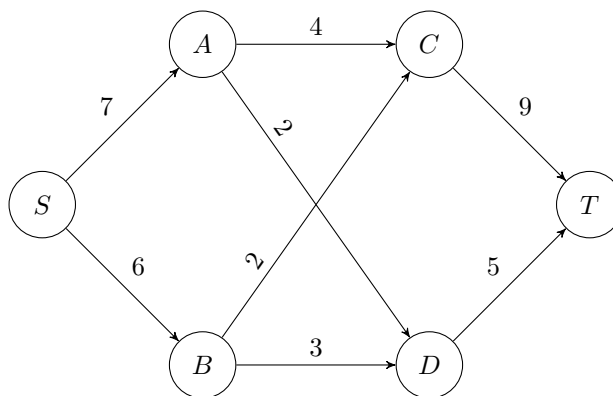


Figure 1: Network

- Find the maximum flow f and a minimum cut.
- Draw the residual graph G_f (along with its edge capacities). In this residual network, mark the vertices reachable from S and the vertices from which T is reachable.

- (c) An edge of a network is called a *bottleneck edge* if increasing its capacity results in an increase in the maximum flow. List all bottleneck edges in the above network.
- (d) Give a very simple example (containing at most four nodes) of a network which has no bottleneck edges.
- (e) Give an efficient algorithm to identify all bottleneck edges in a network (Hint: Start by running the usual network flow algorithm, and then examine the residual graph.)
3. (20 points) Suppose in a particular network $G = (V, E)$ whose edges have integer capacities $c(e)$, we have already found the maximum flow f from vertex s to vertex t . However, we now find out that one of the capacity values we used was wrong: for edge (u, v) we used c_{uv} where it should have been $c_{uv} - 1$. This is unfortunate because the flow f uses that particular edge at full capacity: $f_{uv} = c_{uv}$. We could redo the flow computation from scratch, but there's a faster way. Give an algorithm to compute a new optimal flow in $\mathcal{O}(|V| + |E|)$ time.

Hints: The current flow does not satisfy the constraints. First fix the flow, but note that there might be another path to again increase the flow.

4. (15 points) Decide whether you think the following state is true or false. If it is true, give a short explanation. If it is false give a counter-example. Let G be an arbitrary flow network, with a source s , a sink t , and a positive integers capacity c_e on every edge, and let (A, B) be a minimum $s - t$ cut with respect to these capacities. Now suppose we add 1 to every capacity, then (A, B) is still a minimum $s - t$ cut with respect to these new capacities.
5. (25 points) We want to choose scientific experiments among n experiments $E_1, \dots, E_n$ that must be done in outer space. Each experiment needs tools that should be chosen among p tools $I_1, \dots, I_p$. Each tool may be used by multiple experiments. By conducting the experiment E_i , we earn p_i euros, but carrying the tool I_j into outer space costs c_j euros.

The goal is to maximize the profits.

Let us consider the following graph: for each experiment E_i , we add a vertex E_i . For each tool I_j , we add a vertex I_j . There is also a source s and a sink t . An edge of capacity p_i goes from s to E_i and an edge of capacity c_j goes from I_j to t . Finally, if the experiment E_i uses the tool I_j , then there is an edge of infinite capacity from E_i to I_j .

- (a) What is the graph associated to the following experiments?

	I_1	I_2	I_3	I_4	I_5	I_6
E_1	x	x	x			
E_2	x			x	x	
E_3			x			x
E_4		x		x	x	x

- (b) How can we find the solution (that is to say, the set of experiments) that maximizes the profits? How can we compute the total profits?